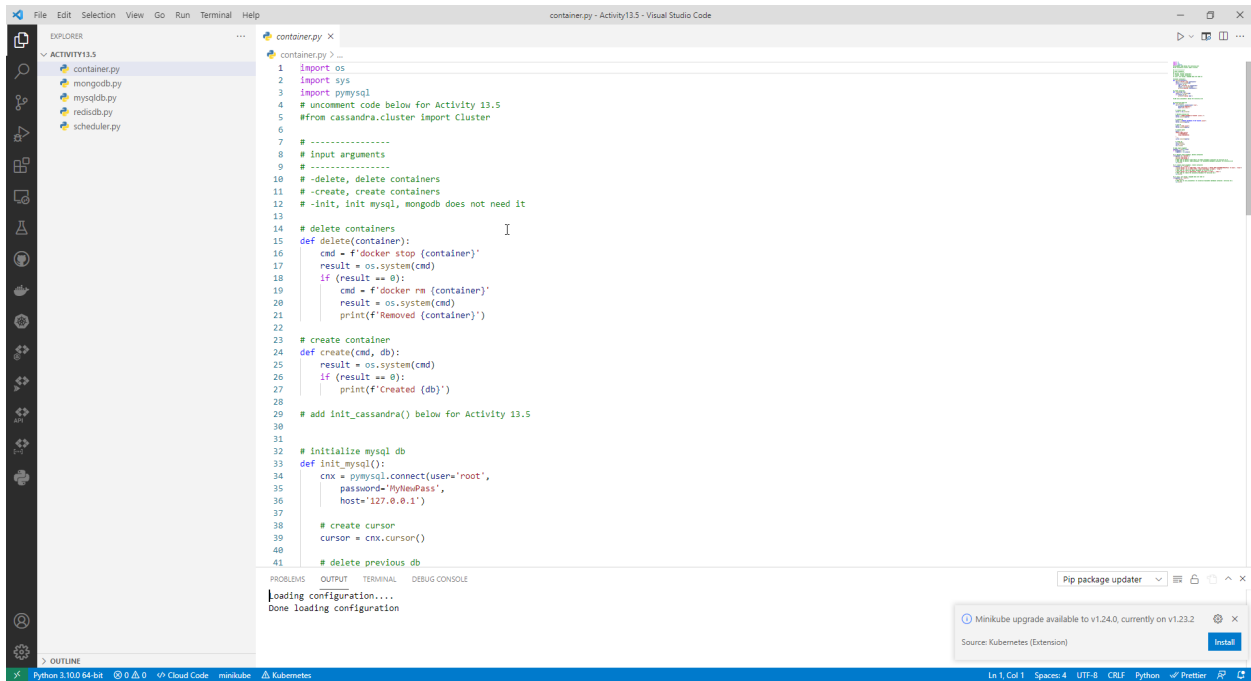


Activity 13.5 Solution File

1.

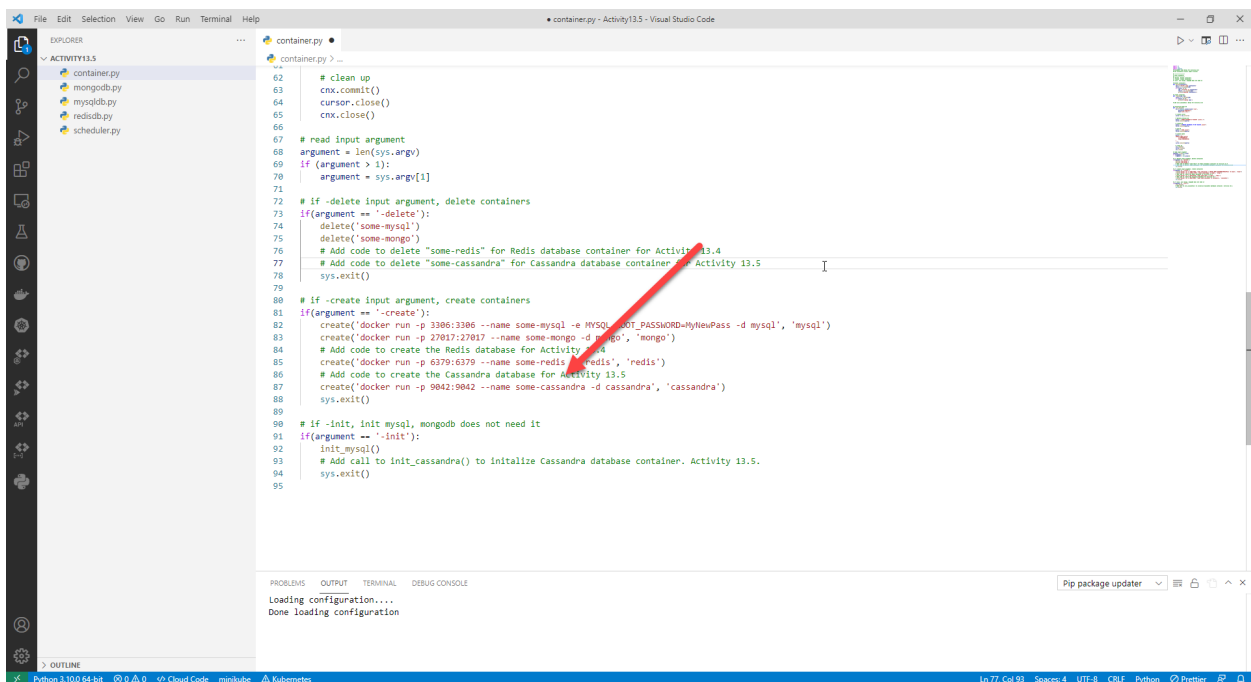


```

1  import os
2  import sys
3  import pymysql
4  # Uncomment code below for Activity 13.5
5  # from cassandra.cluster import Cluster
6
7  # -----
8  # Input arguments
9  # -----
10 # -delete, delete containers
11 # -create, create containers
12 # -init, init mysql, mongodb does not need it
13
14 # delete containers
15 def delete(container):
16     cmd = f'docker stop {container}'
17     result = os.system(cmd)
18     if (result == 0):
19         cmd = f'docker rm {container}'
20         result = os.system(cmd)
21         print(f'Removed {container}')
22
23 # create container
24 def create(cmd, db):
25     result = os.system(cmd)
26     if (result == 0):
27         print(f'Created {db}')
28
29 # Add init_cassandra() below for Activity 13.5
30
31 # initialize mysql db
32 def init_mysql():
33     cnx = pymysql.connect(user='root',
34                           password='MyNewPass',
35                           host='127.0.0.1')
36
37     # create cursor
38     cursor = cnx.cursor()
39
40     # delete previous db
41
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```

2.

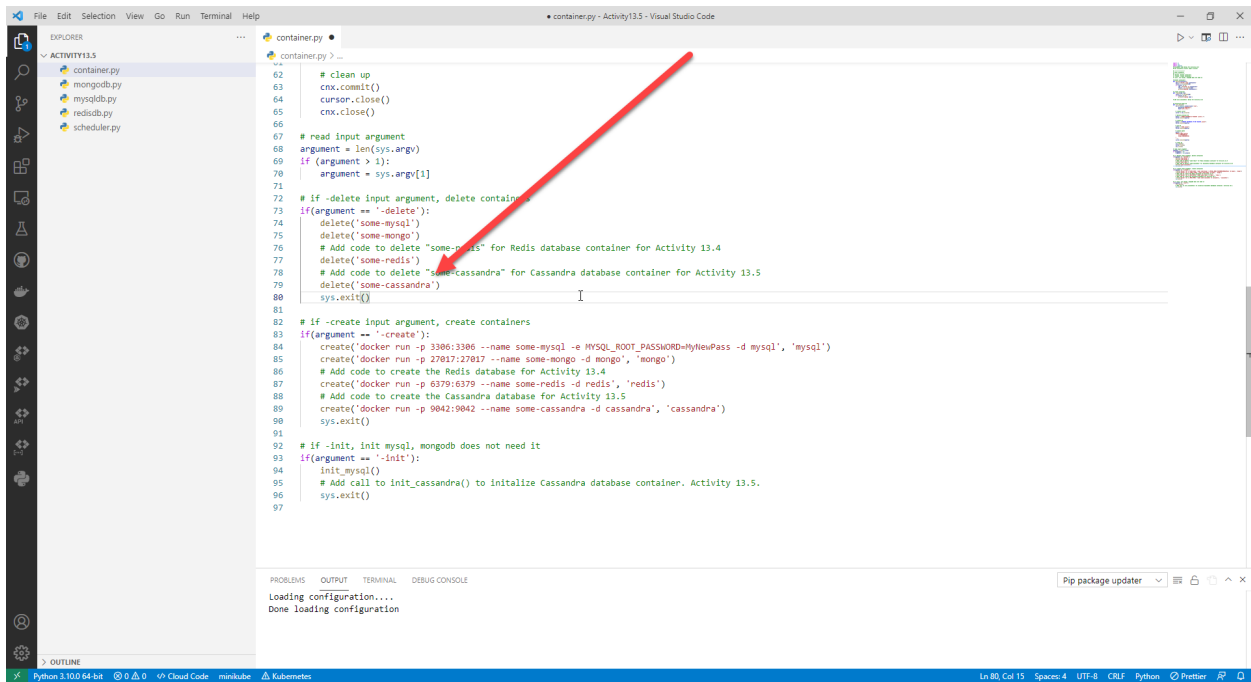


```

62 # clean up
63 cnx.commit()
64 cursor.close()
65 cnx.close()
66
67 # read input argument
68 argument = len(sys.argv)
69 if (argument > 1):
70     argument = sys.argv[1]
71
72 # if -delete input argument, delete containers
73 if (argument == '-delete'):
74     delete('some-mysql')
75     delete('some-mongo')
76     # Add code to delete "some-redis" for Redis database container for Activity 13.4
77     # Add code to delete "some-cassandra" for Cassandra database container for Activity 13.5
78     sys.exit()
79
80 # if -create input argument, create containers
81 if (argument == '-create'):
82     create('docker run -p 3306:3306 --name some-mysql -e MYSQL_ROOT_PASSWORD=MyNewPass -d mysql', 'mysql')
83     create('docker run -p 27017:27017 --name some-mongo -d mongo', 'mongo')
84     # Add code to create the Redis database for Activity 13.4
85     create('docker run -p 6379:6379 --name some-redis -d redis', 'redis')
86     # Add code to create the Cassandra database for Activity 13.5
87     create('docker run -p 9042:9042 --name some-cassandra -d cassandra', 'cassandra')
88     sys.exit()
89
90 # if -init, init mysql, mongodb does not need it
91 if (argument == '-init'):
92     init_mysql()
93     # Add call to init_cassandra() to initialize Cassandra database container. Activity 13.5.
94     sys.exit()
95

```

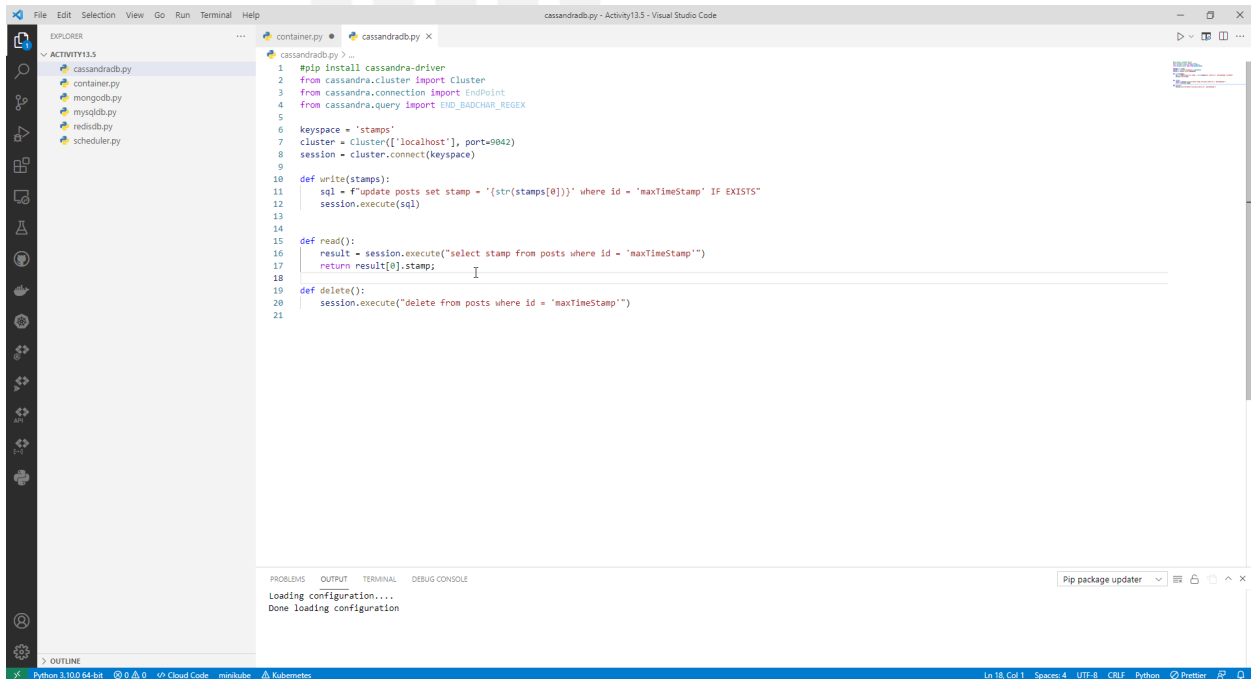
3.



```

62 # clean up
63 cnx.commit()
64 cursor.close()
65 cnx.close()
66
67 # read input argument
68 argument = len(sys.argv)
69 if argument > 1:
70     argument = sys.argv[1]
71
72 # If -delete input argument, delete containers
73 if(argument == '-delete'):
74     delete('some-mysql')
75     delete('some-mongo')
76     # Add code to delete 'some-redis' for Redis database container for Activity 13.4
77     delete('some-redis')
78     # Add code to delete 'some-cassandra' for Cassandra database container for Activity 13.5
79     delete('some-cassandra')
80     sys.exit()
81
82 # If -create input argument, create containers
83 if(argument == '-create'):
84     create('docker run -p 3306:3306 --name some-mysql -e MYSQL_ROOT_PASSWORD=whyNewPass -d mysql', 'mysql')
85     create('docker run -p 27017:27017 --name some-mongo -d mongo', 'mongo')
86     # Add code to create the Redis database for Activity 13.4
87     create('docker run -p 6379:6379 --name some-redis -d redis', 'redis')
88     # Add code to create the Cassandra database for Activity 13.5
89     create('docker run -p 9042:9042 --name some-cassandra -d cassandra', 'cassandra')
90     sys.exit()
91
92 # If -init, init mysql, mongodb does not need it
93 if(argument == '-init'):
94     init_mysql()
95     # Add call to init_cassandra() to initialize Cassandra database container. Activity 13.5.
96     sys.exit()
97
  
```

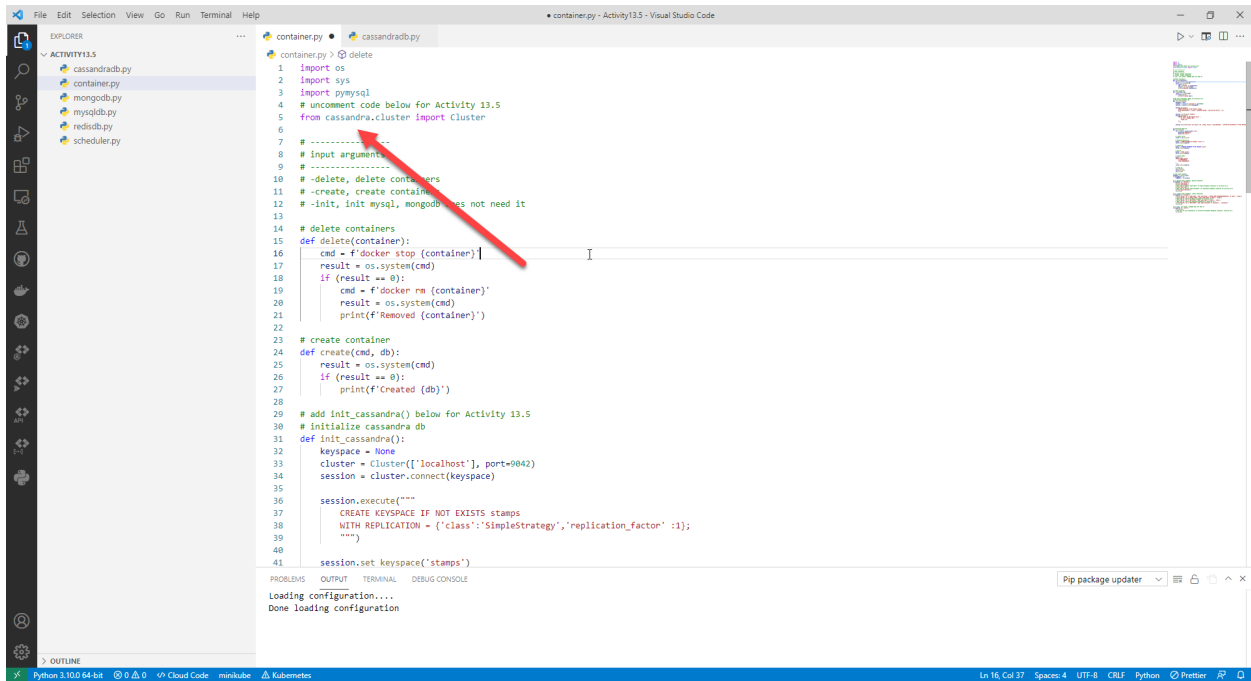
4.



```

1 #pip install cassandra-driver
2 from cassandra.cluster import Cluster
3 from cassandra.connection import EndPoint
4 from cassandra.query import END_BADCHAR_REGEX
5
6 keyspace = 'stamps'
7 cluster = Cluster(['localhost'], port=9042)
8 session = cluster.connect(keyspace)
9
10 def write(stamps):
11     sql = f"update posts set stamp = '{str(stamps[0])}' where id = 'maxTimeStamp' IF EXISTS"
12     session.execute(sql)
13
14
15 def read():
16     result = session.execute("select stamp from posts where id = 'maxTimeStamp'")
17     return result[0].stamp
18
19 def delete():
20     session.execute("delete from posts where id = 'maxTimeStamp'")
21
  
```

5a.

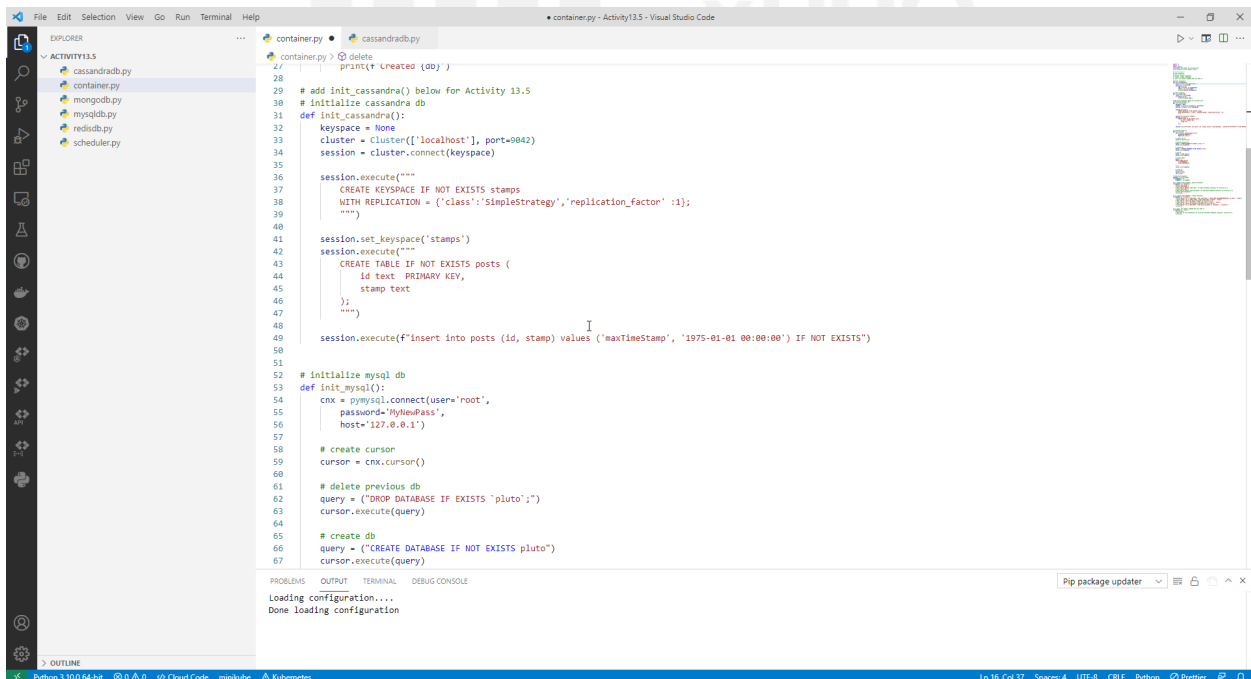


```

1  import os
2  import sys
3  import pymysql
4  # uncomment code below for Activity 13.5
5  from cassandra.cluster import Cluster
6
7  # -----
8  # input arguments
9  # -----
10 # delete, delete containers
11 # -create, create containers
12 # -init, init mysql, mongodb, etc not need it
13
14 # delete containers
15 def delete(container):
16     cmd = f'docker stop {container}'
17     result = os.system(cmd)
18     if (result == 0):
19         cmd = f'docker rm {container}'
20         result = os.system(cmd)
21         print(f'Removed {container}')
22
23 # create container
24 def create(cad, db):
25     result = os.system(cmd)
26     if (result == 0):
27         print(f'Created {db}')
28
29 # add init_cassandra() below for Activity 13.5
30 # initialize cassandra db
31 def init_cassandra():
32     keyspace = None
33     cluster = Cluster(['localhost'], port=9042)
34     session = cluster.connect(keyspace)
35
36     session.execute("""
37         CREATE KEYSPACE IF NOT EXISTS stamps
38         WITH REPLICATION = {'class': 'SimpleStrategy', 'replication_factor': 1};
39     """)
40     session.set_keyspace('stamps')
41
42 # initialize mysql db
43 def init_mysql():
44     cnx = pymysql.connect(user='root',
45                           password='HyNeuPass',
46                           host='127.0.0.1')
47
48     # create cursor
49     cursor = cnx.cursor()
50
51     # delete previous db
52     query = f'DROP DATABASE IF EXISTS `pluto`;`
53     cursor.execute(query)
54
55     # create db
56     query = f'CREATE DATABASE IF NOT EXISTS pluto;`
57     cursor.execute(query)

```

5b.

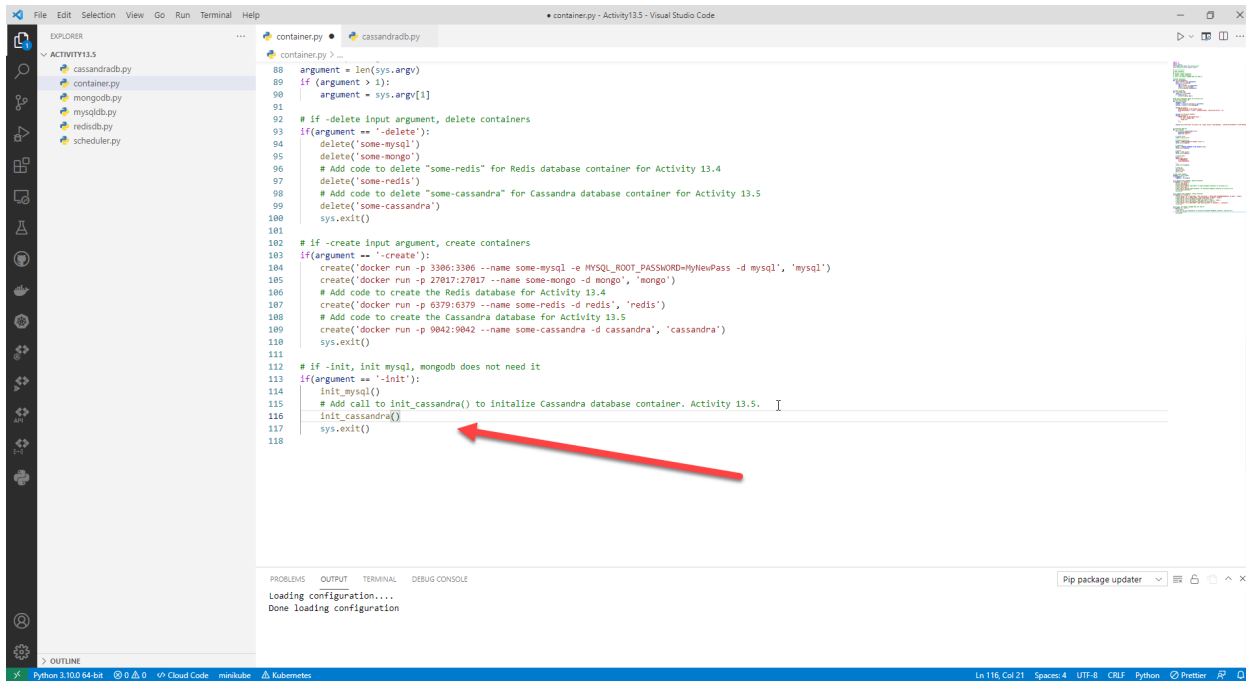


```

27 # print('Created {db}')
28
29 # add init_cassandra() below for Activity 13.5
30 # initialize cassandra db
31 def init_cassandra():
32     keyspace = None
33     cluster = Cluster(['localhost'], port=9042)
34     session = cluster.connect(keyspace)
35
36     session.execute("""
37         CREATE KEYSPACE IF NOT EXISTS stamps
38         WITH REPLICATION = {'class': 'SimpleStrategy', 'replication_factor': 1};
39     """)
40     session.set_keyspace('stamps')
41
42     session.execute("""
43         CREATE TABLE IF NOT EXISTS posts (
44             id text PRIMARY KEY,
45             stamp text
46         );
47     """)
48     session.execute(f'insert into posts (id, stamp) values ('maxTimeStamp', '1975-01-01 00:00:00') IF NOT EXISTS')
49
50 # initialize mysql db
51 def init_mysql():
52     cnx = pymysql.connect(user='root',
53                           password='HyNeuPass',
54                           host='127.0.0.1')
55
56     # create cursor
57     cursor = cnx.cursor()
58
59     # delete previous db
60     query = f'DROP DATABASE IF EXISTS `pluto`;`
61     cursor.execute(query)
62
63     # create db
64     query = f'CREATE DATABASE IF NOT EXISTS pluto;`
65     cursor.execute(query)

```

5c.

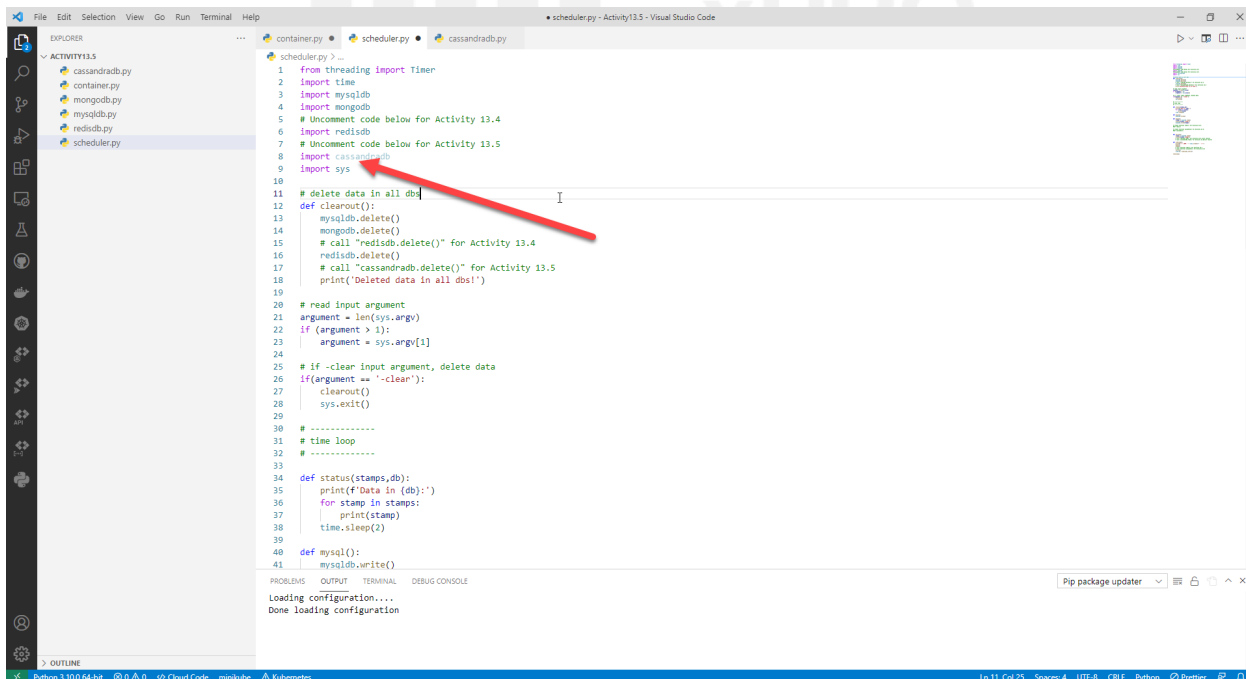


```

88 argument = len(sys.argv)
89 if (argument > 1):
90     argument = sys.argv[1]
91
92 # If -delete input argument, delete containers
93 if (argument == "-delete"):
94     delete('some-mysql')
95     delete('some-mongo')
96     # Add code to delete "some-redis" for Redis database container for Activity 13.4
97     delete('some-redis')
98     # Add code to delete "some-cassandra" for Cassandra database container for Activity 13.5
99     delete('some-cassandra')
100     sys.exit()
101
102 # If -create input argument, create containers
103 if (argument == "-create"):
104     create('docker run -p 3306:3306 --name some-mysql -e MYSQL_ROOT_PASSWORD=HyfieuPass -d mysql', 'mysql')
105     create('docker run -p 27017:27017 --name some-mongo -d mongo', 'mongo')
106     # Add code to create the Redis database for Activity 13.4
107     create('docker run -p 6379:6379 --name some-redis -d redis', 'redis')
108     # Add code to create the Cassandra database for Activity 13.5
109     create('docker run -p 9042:9042 --name some-cassandra -d cassandra', 'cassandra')
110     sys.exit()
111
112 # If -init, init mysql, mongodb does not need it
113 if (argument == "-init"):
114     init_mysql()
115     # Add call to init_cassandra() to initialize Cassandra database container. Activity 13.5.
116     init_cassandra()
117     sys.exit()
118

```

6a.

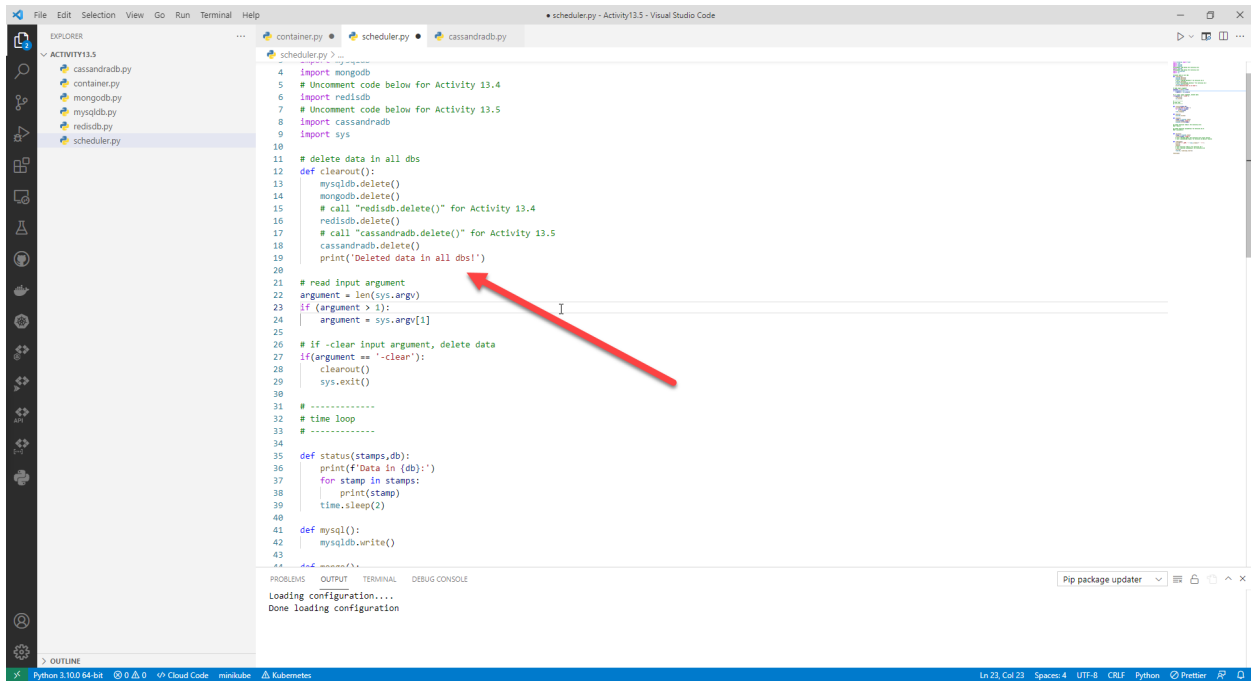


```

1 from threading import Timer
2 import time
3 import mysql
4 import mongo
5 # Uncomment code below for Activity 13.4
6 import redis
7 # Uncomment code below for Activity 13.5
8 import cassandradb
9 import sys
10
11 # delete data in all dbs
12 def clearout():
13     mysql.delete()
14     mongo.delete()
15     # call "redis.delete()" for Activity 13.4
16     redis.delete()
17     # call "cassandradb.delete()" for Activity 13.5
18     print("Deleted data in all dbs")
19
20 # read input argument
21 argument = len(sys.argv)
22 if (argument > 1):
23     argument = sys.argv[1]
24
25 # If -clear input argument, delete data
26 if (argument == "-clear"):
27     clearout()
28     sys.exit()
29
30 # -----
31 # time loop
32 # -----
33
34 def status(stamps,db):
35     print(f'Data in (db):')
36     for stamp in stamps:
37         print(stamp)
38         time.sleep(2)
39
40 def mysql():
41     mysql.write()

```

6b.

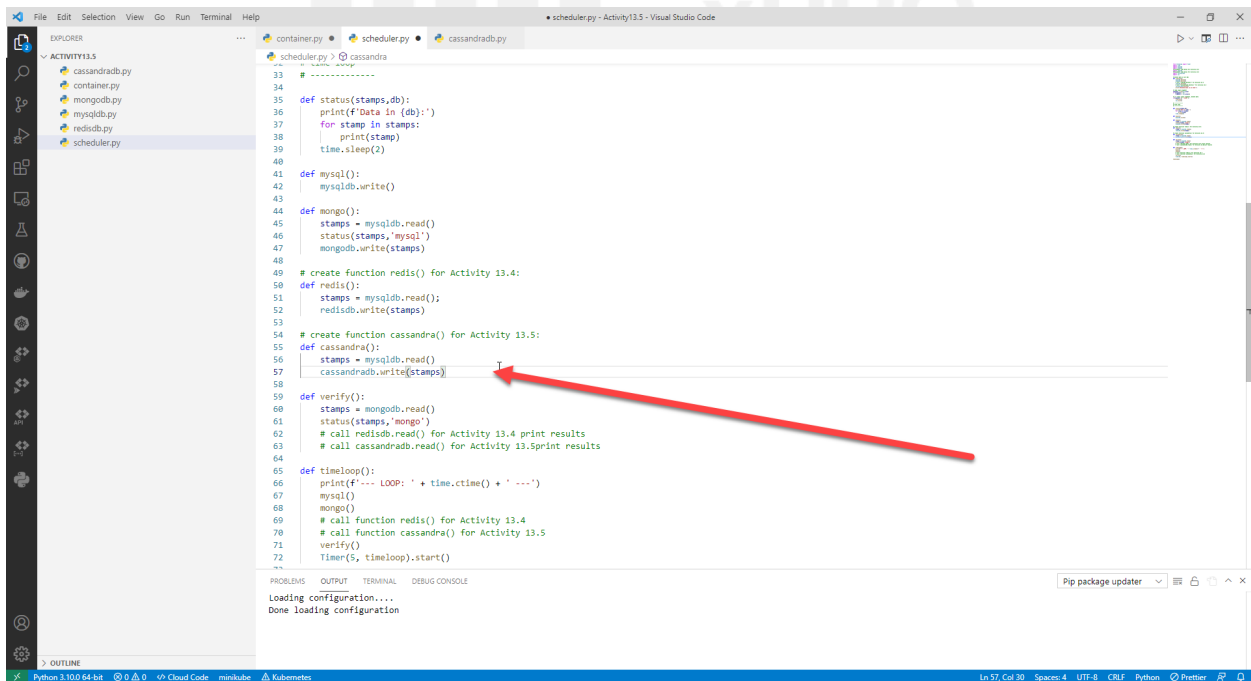


```

4 import mongodb
5 # Uncomment code below for Activity 13.4
6 import redisdb
7 # Uncomment code below for Activity 13.5
8 import cassandradb
9 import sys
10
11 # delete data in all dbs
12 def clearout():
13     mysql.delete()
14     mongodb.delete()
15     # call "redisdb.delete()" for Activity 13.4
16     redisdb.delete()
17     # call "cassandradb.delete()" for Activity 13.5
18     cassandradb.delete()
19     print('Deleted data in all dbs!')
20
21 # read input argument
22 argument = len(sys.argv)
23 if (argument > 1):
24     argument = sys.argv[1]
25
26 # if -clear input argument, delete data
27 if(argument == '-clear'):
28     clearout()
29     sys.exit()
30
31 # -----
32 # time loop
33 # -----
34
35 def status(stamps,db):
36     print(f'Data in {db}:')
37     for stamp in stamps:
38         print(stamp)
39     time.sleep(2)
40
41 def mysql():
42     mysql.write()
43
44
45
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```

6c.

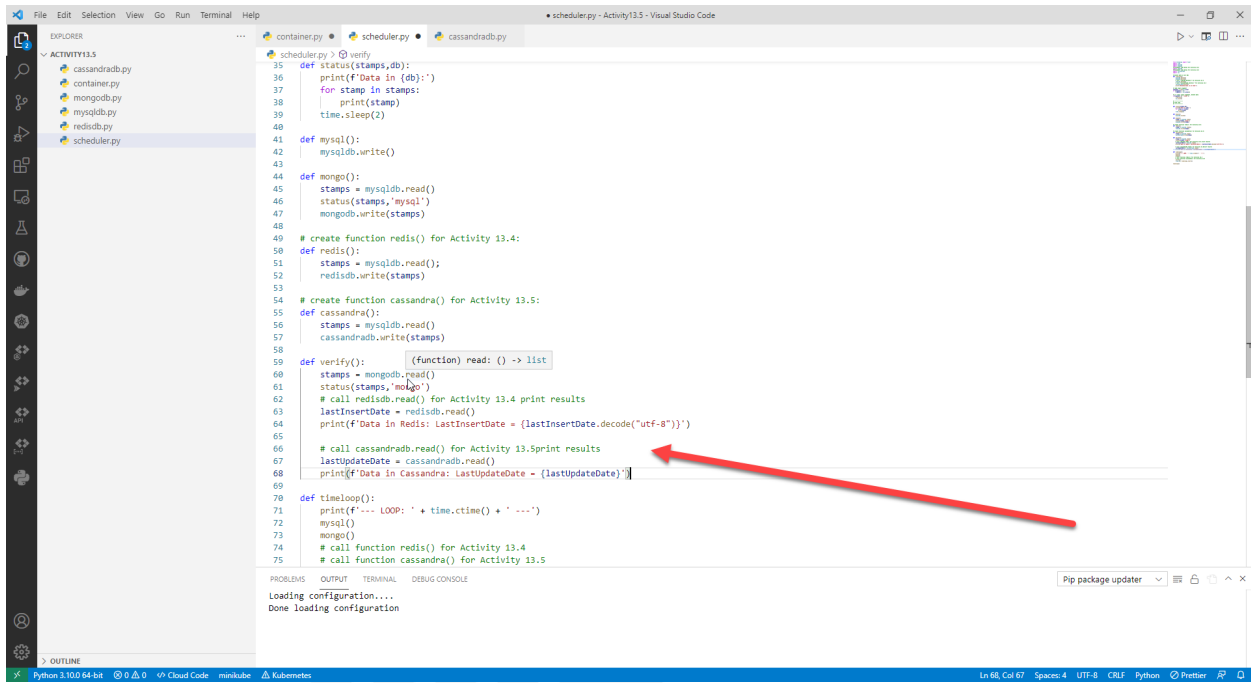


```

33 # -----
34
35 def status(stamps,db):
36     print(f'Data in {db}:')
37     for stamp in stamps:
38         print(stamp)
39     time.sleep(2)
40
41 def mysql():
42     mysql.write()
43
44 def mongo():
45     stamps = mysql.read()
46     status(stamps,'mysql')
47     mongodb.write(stamps)
48
49 # create function redis() for Activity 13.4:
50 def redis():
51     stamps = mysql.read()
52     redisdb.write(stamps)
53
54 # create function cassandra() for Activity 13.5:
55 def cassandra():
56     stamps = mysql.read()
57     cassandradb.write(stamps)
58
59 def verify():
60     stamps = mongodb.read()
61     status(stamps,'mongo')
62     # call redisdb.read() for Activity 13.4 print results
63     # call cassandradb.read() for Activity 13.5 print results
64
65 def timeLoop():
66     print(f'--- LOOP: ' + time.ctime() + ' ---')
67     mysql()
68     mongo()
69     # call function redis() for Activity 13.4
70     # call function cassandra() for Activity 13.5
71     verify()
72     Timer(5, timeLoop).start()
73
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```

6d.

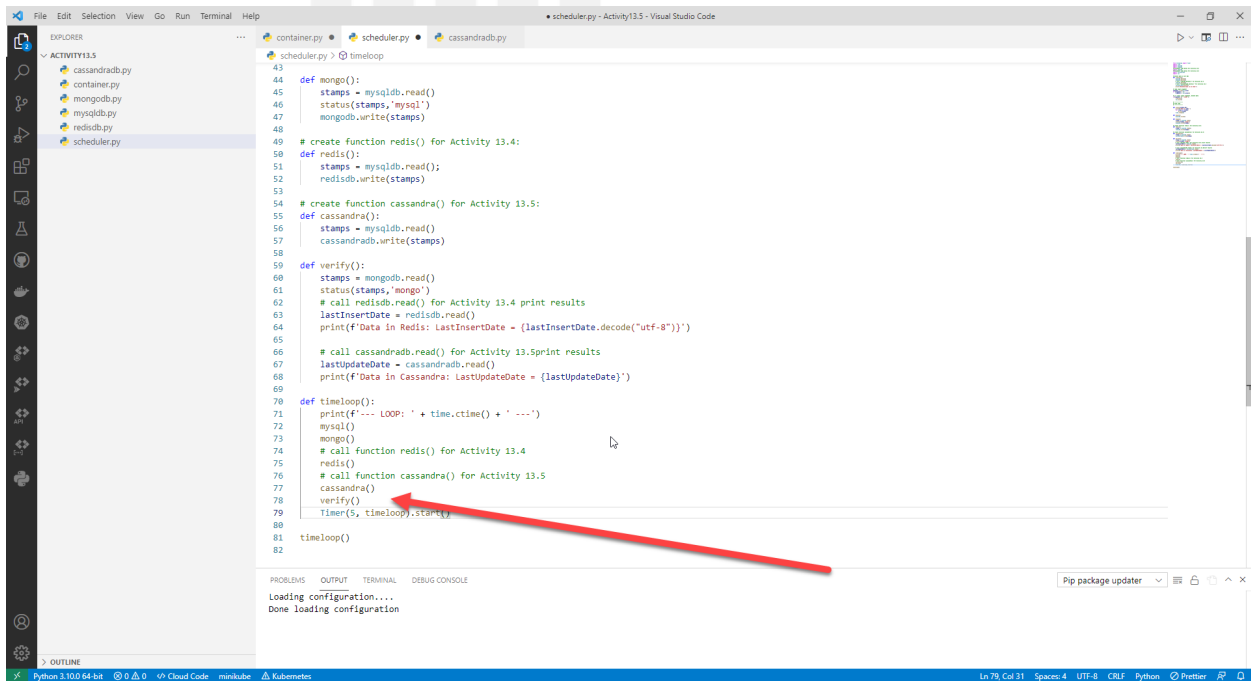


```

35 def status(stamps,db):
36     print(f'Data in (db):')
37     for stamp in stamps:
38         print(stamp)
39         time.sleep(2)
40
41 def mysql():
42     mysqlLib.write()
43
44 def mongo():
45     stamps = mysqlLib.read()
46     status(stamps,'mysql')
47     mongoLib.write(stamps)
48
49 # create function redis() for Activity 13.4:
50 def redis():
51     stamps = mysqlLib.read();
52     redisLib.write(stamps)
53
54 # create function cassandra() for Activity 13.5:
55 def cassandra():
56     stamps = mysqlLib.read()
57     cassandradb.write(stamps)
58
59 def verify():
60     stamps = mongoLib.read()
61     status(stamps,'mongo')
62     # call redisLib.read() for Activity 13.4 print results
63     lastInsertDate = redisLib.read()
64     print(f'Data in Redis: LastInsertDate = {lastInsertDate.decode("utf-8")}')
65
66     # call cassandradb.read() for Activity 13.5 print results
67     lastUpdateDate = cassandradb.read()
68     print(f'Data in Cassandra: LastUpdateDate = {lastUpdateDate}')
69
70 def timeLoop():
71     print(f'--- LOOP: ' + time.ctime() + ' ---')
72     mysql()
73     mongo()
74     # call function redis() for Activity 13.4
75     # call function cassandra() for Activity 13.5

```

6e.



```

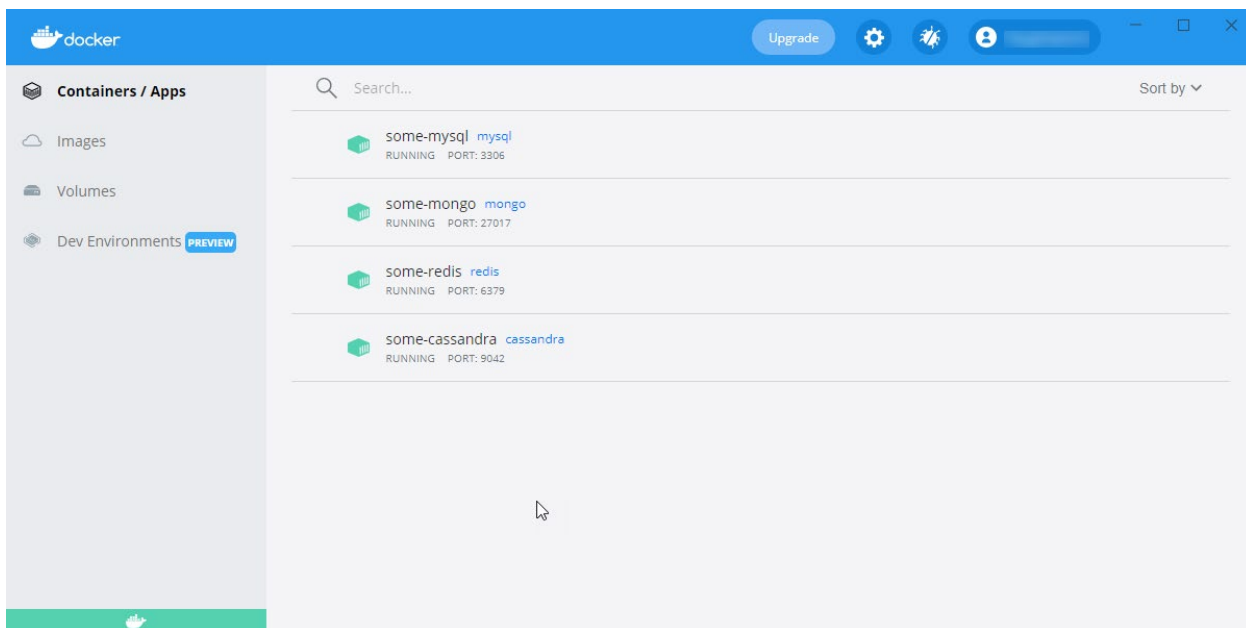
43
44 def mongo():
45     stamps = mysqlLib.read()
46     status(stamps,'mysql')
47     mongoLib.write(stamps)
48
49 # create function redis() for Activity 13.4:
50 def redis():
51     stamps = mysqlLib.read();
52     redisLib.write(stamps)
53
54 # create function cassandra() for Activity 13.5:
55 def cassandra():
56     stamps = mysqlLib.read()
57     cassandradb.write(stamps)
58
59 def verify():
60     stamps = mongoLib.read()
61     status(stamps,'mongo')
62     # call redisLib.read() for Activity 13.4 print results
63     lastInsertDate = redisLib.read()
64     print(f'Data in Redis: LastInsertDate = {lastInsertDate.decode("utf-8")}')
65
66     # call cassandradb.read() for Activity 13.5 print results
67     lastUpdateDate = cassandradb.read()
68     print(f'Data in Cassandra: LastUpdateDate = {lastUpdateDate}')
69
70 def timeLoop():
71     print(f'--- LOOP: ' + time.ctime() + ' ---')
72     mysql()
73     mongo()
74     # call function redis() for Activity 13.4
75     redis()
76     # call function cassandra() for Activity 13.5
77     cassandra()
78     verify()
79     Timer(5, timeLoop, status)
80
81 timeLoop()
82

```

7.

```
Windows PowerShell
PS D:\MO-PCDE\Module13\Activity13.5> python container.py -create
1fb0204b2800b6ce5d6c665680eae83a10524fe91955afc511b933532a39ae3
Created mysql
3feba4a5288293e96f9ceeeaa1dab8858e5e326bdb40af66581a4d732ae99ce05
Created mongo
190b808e1a89085901bf4dee2cad29139b087362153df5d81b96b9fe9053d1bd
Created redis
3ad26edfbd721527ca19ed69bb2bdb03cc6052591d7e3e141e78665dbbb5c7c
Created cassandra
PS D:\MO-PCDE\Module13\Activity13.5>
```

8.



9.

```
Windows PowerShell
PS D:\MO-PCDE\Module13\Activity13.5> python container.py -init
PS D:\MO-PCDE\Module13\Activity13.5>
```

10.

```

Windows PowerShell
PS D:\MO-PCDE\Module13\Activity13.5> python scheduler.py
--- LOOP: Sun Nov  7 08:42:19 2021 ---
Data in mysql:
2021-11-07 08:42:19
2021-11-07 08:28:43
2021-11-07 08:28:34
Data in mongo:
2021-11-07 08:42:19
2021-11-07 08:28:43
2021-11-07 08:28:34
Data in Redis: LastInsertDate = 2021-11-07 08:42:19
Data in Cassandra: LastUpdateDate = 2021-11-07 08:42:19
--- LOOP: Sun Nov  7 08:42:28 2021 ---
Data in mysql:
2021-11-07 08:42:28
2021-11-07 08:42:19
2021-11-07 08:28:43
2021-11-07 08:28:34
Data in mongo:
2021-11-07 08:42:28
2021-11-07 08:42:19
2021-11-07 08:28:43
2021-11-07 08:28:34
Data in Redis: LastInsertDate = 2021-11-07 08:42:28
Data in Cassandra: LastUpdateDate = 2021-11-07 08:42:28
--- LOOP: Sun Nov  7 08:42:37 2021 ---
Data in mysql:
2021-11-07 08:42:37
2021-11-07 08:42:28
2021-11-07 08:42:19
2021-11-07 08:28:43
2021-11-07 08:28:34
Data in mongo:
2021-11-07 08:42:37
2021-11-07 08:42:28
2021-11-07 08:42:19
2021-11-07 08:28:43
2021-11-07 08:28:34
Data in Redis: LastInsertDate = 2021-11-07 08:42:37
Data in Cassandra: LastUpdateDate = 2021-11-07 08:42:37

```

11.

```

Windows PowerShell
PS D:\MO-PCDE\Module13\Activity13.5> python container.py -delete
some-mysql
some-mysql
Removed some-mysql
some-mongo
some-mongo
Removed some-mongo
some-redis
some-redis
Removed some-redis
some-cassandra
some-cassandra
Removed some-cassandra
PS D:\MO-PCDE\Module13\Activity13.5>

```


12.

